



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to USHA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

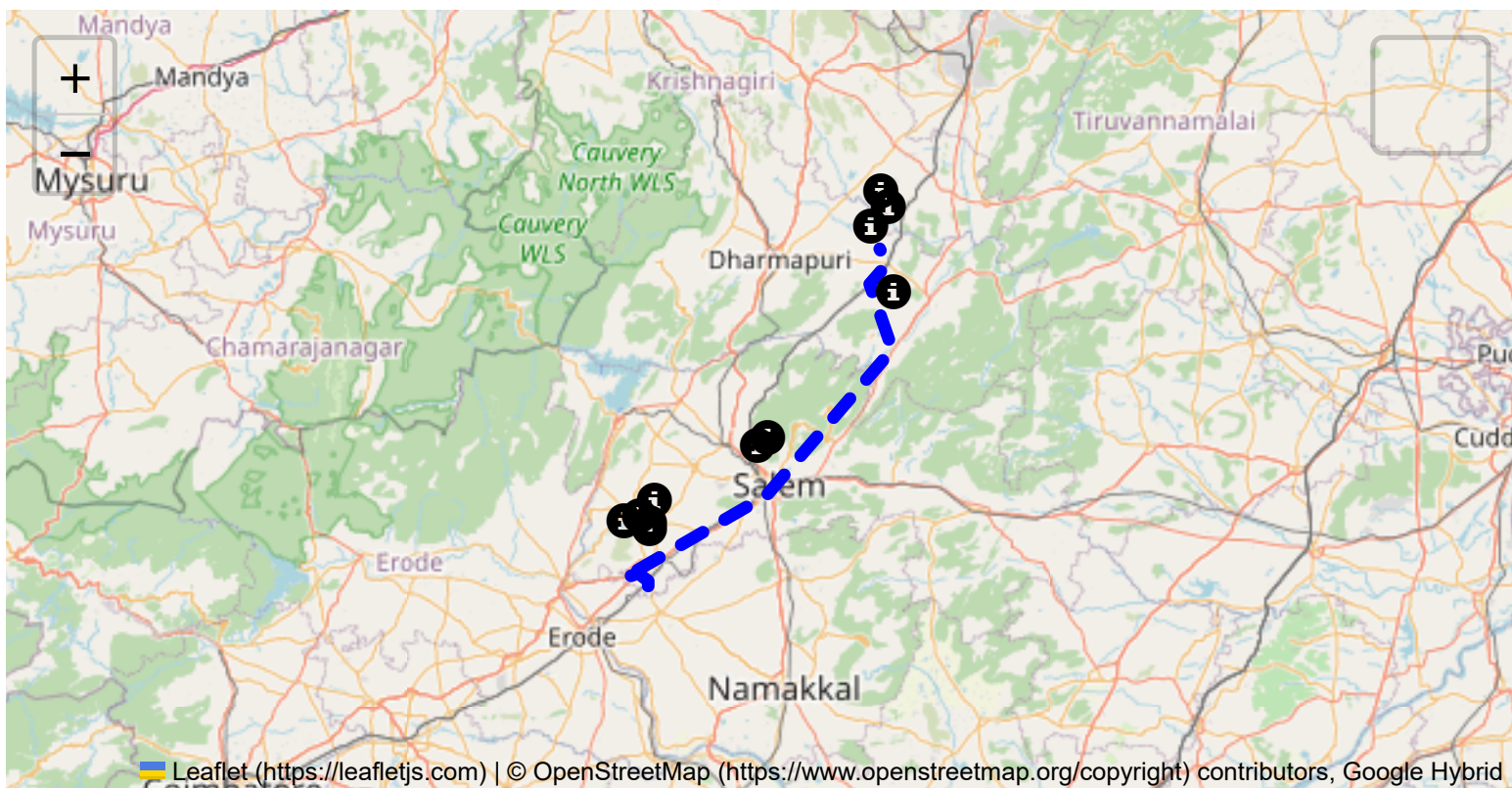
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 130.46 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.159343, 78.3749856)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route covers a significant stretch of road in Tamil Nadu, India, starting from Sangagiri and ending in Chinnakoundampatti. It traverses various regional roads and the NH544 highway, linking several towns such as Sankari, Salem, and Pallipatti. The journey spans approximately 130.46 kilometers, with diverse

environments ranging from urban settings to rural landscapes. The predominant roads are national highways and local main roads, known for their varying width and traffic conditions.

2. Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. The period between March and June can be extremely hot, while the monsoon seasons from July to September and November to December can cause heavy rainfall. Drivers should be vigilant for potential flood-prone areas especially in low-lying regions during monsoons, which can make roads slippery and reduce visibility.

3. Traffic Patterns

Traffic congestion is common around Salem, particularly during peak hours in the morning (8 AM - 10 AM) and evening (5 PM - 8 PM). The NH544 and major junctions in Salem and Sankari are known for their heavy traffic. Drivers should be prepared for slower moving traffic in these areas and plan to travel during off-peak hours if possible.

4. Assessment of Road Quality

The NH544 is generally maintained well, with decent road quality and clear signage. However, some local roads, especially in rural sections, may be narrower with varying road surface quality. There may be occasional potholes and uneven patches in less maintained areas, requiring careful navigation, especially for heavy vehicles.

5. Alternative Routes for Emergencies

In case of emergencies, consider leveraging alternative routes such as NH44 via Dharmapuri if access to NH544 is compromised. Local roads through Tiruchengode and nearby towns may also serve as detours, though these might add extra travel time due to narrower lanes and local traffic.

6. Local Regulations on Hazardous Material Transport

Local regulations require that trucks carrying hazardous materials adhere to specific safety standards, including permissible operation times to minimize risk. Restrictions on the transport of specific hazardous substances may apply. Compliance with Tamil Nadu's road safety and hazardous material regulations is crucial.

7. Historical Incidents

While no specific historical data is readily available for each segment, incidents involving heavy vehicles occasionally occur on highways due to high speeds and dense traffic. NH544, being a major transport artery, has seen its share of vehicular incidents, emphasizing the need for caution.

8. Environmental Considerations

Environmental sensitivity is particularly important near water bodies and protected areas. Along this route, drivers need to be aware of any nearby agricultural land or ecological zones that could be affected

by spillage or accidents involving hazardous materials.

9. Communication Coverage

Broad communication coverage is generally good along highways but may weaken in rural regions and certain pockets between towns. Drivers need to ensure that communication devices are fully charged and equipped with service providers offering the best rural coverage.

10. Estimated Emergency Response Times

Emergency response times can vary significantly: urban areas like Salem might see responses within 15-30 minutes, while rural areas might experience delays up to an hour or more due to distance from emergency facilities.

11. Overall Risk Assessment Summary

Assessing the entire journey from Sangagiri to Chinnakoundampatti, the risk factors primarily include adverse weather, traffic congestion, and road quality variability. High caution is advised during peak hours and inclement weather. Alternative routes should be preplanned for emergencies, ensuring all local regulations are adhered to for hazardous material transport. Comprehensive pre-trip inspections and understanding environmental and communication factors can mitigate these risks effectively. Overall, with proper planning and caution, this route can be navigated safely by truck drivers transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	High	11.45571, 77.81702	15 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Turn	Medium	11.46916, 77.82431	30 KM/Hr
10	Turn	Medium	11.53560, 77.95773	30 KM/Hr
11	Turn	Medium	11.62100, 78.10723	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Blind Spot	Blind Spot	11.62130, 78.10776	10 KM/Hr
13	Turn	High	11.61249, 78.10443	15 KM/Hr
14	Turn	High	11.61224, 78.10469	15 KM/Hr
15	Turn	Medium	11.62901, 78.16571	30 KM/Hr
16	Turn	Medium	11.65889, 78.20502	30 KM/Hr
17	Turn	Medium	11.66354, 78.20645	30 KM/Hr
18	Turn	Medium	11.67130, 78.23006	30 KM/Hr
19	Blind Spot	Blind Spot	11.67120, 78.23136	10 KM/Hr
20	Turn	High	11.67171, 78.23139	15 KM/Hr
21	Turn	Medium	11.80458, 78.33335	30 KM/Hr
22	Turn	Medium	11.81249, 78.34088	30 KM/Hr
23	Turn	Medium	11.83492, 78.35299	30 KM/Hr
24	Turn	Medium	11.84761, 78.36080	30 KM/Hr
25	Turn	High	11.93919, 78.40422	15 KM/Hr
26	Turn	Medium	11.93992, 78.40355	30 KM/Hr
27	Turn	High	11.94056, 78.40185	15 KM/Hr
28	Turn	Medium	11.95589, 78.39972	30 KM/Hr
29	Turn	Medium	11.96475, 78.39648	30 KM/Hr
30	Turn	Medium	12.00574, 78.39736	30 KM/Hr
31	Turn	Medium	12.00993, 78.39696	30 KM/Hr
32	Turn	Medium	12.01166, 78.39848	30 KM/Hr
33	Turn	Medium	12.01396, 78.39882	30 KM/Hr
34	Turn	Medium	12.01559, 78.39604	30 KM/Hr
35	Turn	Medium	12.06704, 78.36322	30 KM/Hr
36	Turn	Medium	12.07739, 78.35549	30 KM/Hr
37	Turn	High	12.07974, 78.35572	15 KM/Hr
38	Turn	High	12.09483, 78.37653	15 KM/Hr
39	Turn	Medium	12.10614, 78.38344	30 KM/Hr
40	Turn	High	12.10659, 78.38344	15 KM/Hr
41	Turn	High	12.10716, 78.38254	15 KM/Hr
42	Turn	Medium	12.12102, 78.39077	30 KM/Hr
43	Turn	Medium	12.12617, 78.39336	30 KM/Hr
44	Turn	High	12.12757, 78.39470	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
7	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
12	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
14	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
15	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
16	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
17	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
18	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
22	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
6	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
10	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
28	school	Viswa Bharathi	12.1372283, 78.3871787	30 km/h	Medium
29	school	Kongu Matric Hr sec School	12.1398117, 78.3892949	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53560, 77.95773



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62100, 78.10723



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61249, 78.10443



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61224, 78.10469



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62901, 78.16571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65889, 78.20502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66354, 78.20645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67130, 78.23006



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67171, 78.23139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80458, 78.33335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81249, 78.34088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83492, 78.35299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84761, 78.36080



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93919, 78.40422



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93992, 78.40355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.94056, 78.40185



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95589, 78.39972



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96475, 78.39648



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00574, 78.39736



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00993, 78.39696



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01166, 78.39848



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01396, 78.39882



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01559, 78.39604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06704, 78.36322



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07739, 78.35549



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.07974, 78.35572



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09483, 78.37653



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10614, 78.38344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10659, 78.38344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10716, 78.38254



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12102, 78.39077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12617, 78.39336



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.12757, 78.39470